GAT-CTDE: A Correctness-Aware Graph-Attention Architecture for Federated Multi-Agent Slice Admission Control

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Abstract

This paper delivers a working realisation of a graph-attention, multi-agent, federated slice admission controller, GAT-CTDE, treating correctness-aware evaluation as a first-class contribution alongside the architecture, closing three practice gaps in the prior art that this paper's own systematic review identifies (Section 2). On the metric that reflects the actual training objective, GAT-CTDE decisively beats an independent-learner ablation, though its edge over a single-agent baseline is small and not statistically decisive at this sample size, an honest limit we report rather than obscure. Scaled from 3 to 19 gNodeBs (gNBs, 5G base stations) under a disclosed, harder-than-live stress regime, GAT-CTDE resists a training collapse that becomes total for the single-agent baseline, converging, across 64 independently-drawn seeds, on a residual 35.4% [25.5%, 45.8%] collapse rate already present at 7 gNBs. Federation costs nothing measurable in reward. Differential privacy degrades decision precision only past a real threshold, and disruption cost accelerates with severity for every arm. Two safeguards make these results trustworthy: a correctness-aware metric pair (44 of 49 independent checks disagreeing with the standard compliance metric) and a reproduction-versus-replication discipline that caught and retracted one architectural overclaim before publication. A live single-gNB deployment confirms the offline-trained decision logic executes correctly on real OpenAirInterface (OAI) hardware, and surfaces a cross-layer integrity finding: a scheduler-configuration bug had manufactured the appearance of contention in earlier live data, and once fixed, this rig cannot reach genuine per-slice scarcity at any tested load, so the live result establishes deployability rather than admission-control behaviour under contention, tested instead in the disclosed offline stress regime.

Keywords: Network Slicing, Slice Admission Control, Multi-Agent Reinforcement Learning, Graph Attention Networks, Federated Learning, Open RAN

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1 Introduction

Network slicing lets a single physical radio access network serve heterogeneous service classes, namely Ultra-Reliable Low-Latency Communication (URLLC), enhanced Mobile Broadband (eMBB), and massive Machine-Type Communication (mMTC), under distinct per-slice Service Level Agreements (SLAs). Our earlier work [7, 8] established that a Deep Q-Network (DQN) admission controller can learn to enforce these SLAs under a single gNodeB (gNB), validated on a live OpenAirInterface (OAI) testbed. A real multi-cell deployment does not admit requests at one gNB in isolation, since neighbouring cells share backhaul, spectrum planning, and a cluster-wide load-balance objective, making coordinated admission a genuinely multi-agent problem. This raises a direct question: does a shared, graph-structured representation of the cluster's state help a set of per-gNB agents make better joint decisions than a single agent using the flattened joint state, or independent per-gNB agents with no shared representation at all?

This paper's contribution is GAT-CTDE, a centralised-training, decentralised-execution multi-agent architecture built on a shared Graph Attention Network (GAT) [17] encoder over the gNB topology feeding decentralised per-slice Q-heads, delivered together with the evaluation discipline needed to trust what it reports. Three gaps motivate this pairing, each named directly against our own systematic review of 67 core studies (Section 2). First, no reviewed study checks whether its own outcome metric can score a degenerate policy no worse than a correct one. Second, none replicates a headline architectural finding on an independent seed sample before reporting it. Third, none tests whether a live testbed can actually reach the contention regime its offline claims are evaluated under. We close all three: a per-component realisation of the architecture (GAT encoder, Multi-Agent Reinforcement Learning (MARL) controller, Federated Learning (FL) coordinator, Differential Privacy (DP) layer), each with its own measured, isolated contribution rather than one end-to-end number, a correctness-aware metric pair and reproduction-versus-replication discipline applied throughout rather than asserted once, and a live single-gNB deployment that tests, rather than assumes, whether this rig's own achievable load ever produces genuine per-slice scarcity.

Extended to federated training with differential privacy and evaluated under mid-episode gNB dropout, demand spikes, and agent churn, we scale the architecture to cluster sizes $N \in \{7, 19\}$ under ring and hex-grid adjacency. It resists a training collapse that becomes total for a flattened-state single-agent baseline, converging on a residual 35.4% collapse rate across 64 independently-drawn seeds rather than either of two disagreeing small-sample estimates, a pattern already present at $N{=}7$ rather than only emerging at large cluster size. On the metric that reflects the actual training objective, the architecture decisively beats an independent-learner ablation that isolates the same coordination contribution, though its edge over the single-agent baseline specifically is small and not statistically decisive at this sample size. This is an

honest limit we report rather than smooth over to fit a cleaner architecture story. Federating the architecture costs nothing measurable in reward, and differential privacy degrades decision precision only past a real threshold rather than a smooth trade-off. Under disruption, its cost accelerates with severity like every arm tested, with a coordination-specific penalty at only the mildest severity tested.

These safeguards are not abstract. The correctness-aware pair diagnosed and fixed a real training collapse our first encoder suffered on all 30 campaign seeds (Section 4.1), and the reproduction-versus-replication discipline caught and retracted a real architectural overclaim under mid-episode disruption before it could be reported (Section 5.3). The live deployment below both confirms the offline-trained decision logic executes correctly on real hardware and surfaces a cross-layer scheduler misconfiguration that had earlier manufactured the appearance of live contention, fixed at its root, after which this rig cannot exercise genuine per-slice scarcity at any load level tested, confining the disruption and scaling claims above to the disclosed offline stress regime where they are made.

2 Related Work

Our own systematic review [9] screened 6,286 records down to 67 core 2024–2026 studies on machine-learning-driven, user-centric distributed network slicing, organised around three enabling pillars, namely topology-aware graph representation, distributed multi-agent control, and privacy-preserving federation, and found individual pillars well developed but rarely combined under realistic constraints, and never with a validated live deployment. None of the works we reviewed, there or here, checks whether its own evaluation metric can mask a degenerate policy, or replicates a headline finding on an independent seed sample. This is the gap this paper fills.

The closest prior art clusters into three groups against these two checks. Shao et al. [14] (the precedent behind our own GAT encoder choice, showing it strengthens inter-agent cooperation over plain DQN/Advantage Actor-Critic (A2C) baselines), Ahmadi et al. [2] (topology generalisation via graph attention, complementary to, not answered by, the fixed-topology evaluation-integrity question we ask in Section 5.4), and Sulaiman et al. [15] (the closest prior art to our own joint slicing-and-admission-control problem) together check neither whether their own reward-optimal policy could be mistaken for a degenerate one nor a headline result against an independent seed sample. Tashman and Cherkaoui [16] and Fatehi et al. [5] study robustness to a disruptive condition, as our own disruption campaign (Section 5.3) does, but neither checks whether the metric reporting that robustness is itself trustworthy or replicates a finding on an independent sample, exactly the gap our own campaign's independent-seed replication closes, and not hypothetically, since it is what overturned our initial churn-immunity reading. Zhang et al. [19] (the closest periodic-synchronisation precedent) and Yasin et al. [18] (the closest FL+DP precedent, targeting a different threat model, securing the controller rather than characterising a control policy's own privacy-utility trade-off, which our own σ-sweep in Section 5.2 reports) both lack a correctness-aware check on whether federating or privatising changed decision quality. A further, closely related set [3, 6, 12, 13] combines two of these three pillars but, like every work above,

evaluates exclusively on its own reported outcome metric, computed once, on one seed or run. This is the gap this paper fills, not a new point in the GAT/MARL/FL design space but a demonstration (on GAT-CTDE, extended to federation, privacy, and disruption) that checking a metric's own trustworthiness and replicating a finding on an independent sample are both necessary, and that skipping either changes what a reader is entitled to believe the resulting numbers mean.

3 System Model and Problem Formulation

We consider a cluster of N=3 gNBs, each serving the same three slices (URLLC, eMBB, mMTC) under a shared per-gNB Physical Resource Block (PRB) budget B. An admission decision is realised as a PRB-ceiling adjustment (`slicing_control_m` min/max ratio), not a literal per-flow accept/reject, since the real OAI xApp control API only exposes ratio ceilings. No real multi-gNB physical topology is available to us, so we take the least assumption-laden choice and treat the cluster as fully connected, noting this explicitly as a modelling choice rather than a measured fact. At each control step, every gNB has zero or more pending admission requests, each tagged with a slice identity, and an agent decides accept/reject for each independently, conditioned on the current cluster state. The discrete action space matches our single-gNB work exactly (`action_dim=2`), so any coordination benefit found here is attributable to the architecture, not a richer action space.

The reward combines a per-slice SLA-compliance term with a cluster-wide load-balance penalty, weighted by a per-slice `priority_weight` (URLLC highest, mMTC lowest) and a congestion coefficient tuned, in prior work, so that the reward-optimal policy under congestion is *differentiated shedding*: continue accepting URLLC, mostly accept eMBB, and reject mMTC. This is confirmed directly by Q-value inspection on a working baseline policy (Section 4.1), not assumed. Every result in this paper except Section 5.5's bounded live anchor is obtained in an offline, closed-loop simulation environment deliberately oversubscribed to roughly 110% of one gNB's PRB budget, reusing our existing multi-gNB configuration unmodified. A separate investigation found this offline environment's demand process does not reproduce the live testbed's SLA-margin distribution closely enough to support a live-rank prediction claim, even after recalibrating the demand model's temporal structure. A live re-measurement closes this further: at every native load level this rig has been tested at, one and two User Equipment (UEs) per slice alike, all three slices serve essentially 100% of offered demand with zero loss (Section 5.5), meaning this rig has never been shown to reach a load at which any slice's real admission decision carries a genuine capacity consequence. We therefore use this offline environment explicitly as a deliberately harder, disclosed *stress environment* for a contention regime this live rig cannot currently produce, not as a miscalibrated approximation of a reachable live condition. Multi-gNB and disruption claims in this paper remain offline-only throughout, and this rig cannot yet sustain a live multi-gNB test to check them against.

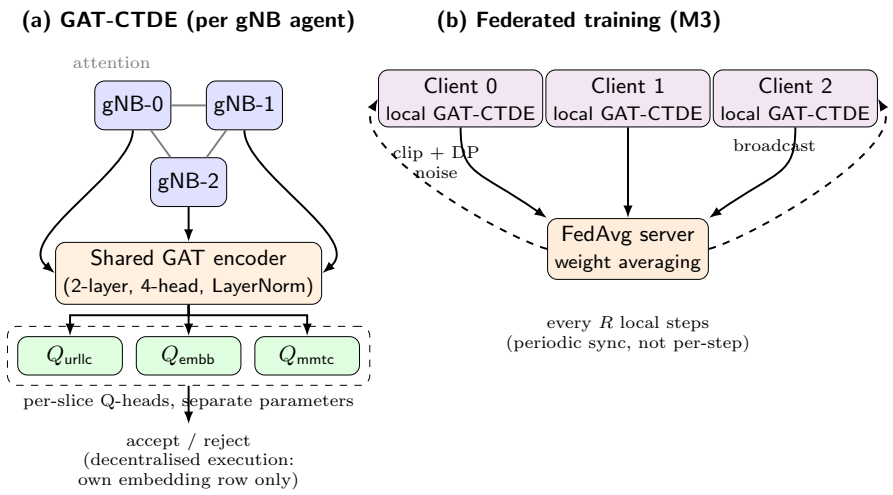


Fig. 1 GAT-CTDE architecture. (a) A shared Graph Attention encoder over the fully-connected 3-gNB topology produces one embedding per node. Per-slice Q-heads, each with its own parameters, consume that node's own embedding row at action-selection time (decentralised execution). (b) The federated training variant: each gNB is a client with a private copy of the same network, synchronised via periodic FedAvg with an optional per-client DP-SGD clip-and-noise step.

4 GAT-CTDE Method

4.1 Shared Encoder, Decentralised Q-Heads

Fig. 1(a) shows the architecture. At every control step, each gNB's own per-slice [PRB-used-ratio, congestion-level, queue-length-norm] features form that node's row in a joint node-feature matrix. A two-layer, four-head GAT [17], implemented directly (the small graph sizes here make a dense, masked-attention implementation as fast as a sparse one), followed by a per-layer `LayerNorm` constraining every node's embedding to zero-mean, unit-variance, produces one embedding per gNB. Centralisation comes from this encoder, since its parameters are updated by every agent's loss jointly, so it sees the full cluster's state during training. Decentralised execution comes from what happens next. At action-selection time, each agent consumes only its own node's embedding row to decide its own pending requests.

Both the `LayerNorm` above and the per-slice heads below were made necessary by a real training collapse, not added on theoretical grounds alone. While generating a federated privacy-utility sweep (Section 5.2), noise levels that should have produced different, stochastically-trained checkpoints instead produced bit-identical per-seed compliance, tracing to our first GAT-CTDE encoder's eval policy blocking zero requests on all 30 campaign seeds. This was a genuine failure, confirmed by a Q-value probe showing $Q(\text{accept}) - Q(\text{reject})$ positive and large for every slice including mMTC, while the independent-DQN ablation (Section 5.1), with no encoder at all, does not collapse. The failure isolated the encoder as the fault: it embedded congestion almost entirely as embedding *magnitude* rather than direction, so that the (roughly linear) downstream Q-head pushed $Q(\text{accept})$ up faster than $Q(\text{reject})$ as congestion

increased, running backwards from the reward's intent. Adding the `LayerNorm` and replacing a single shared Q-head with the per-slice heads described next progressively reduced the collapse from 30/30 to 27/30 to a final 9/30 seeds, with genuine, correctly-targeted differentiated shedding rising correspondingly from 0/30 to 3/30 to 21/30 of the campaign seeds.

An early version fed each embedding, concatenated with a one-hot slice context, into a single shared Q-head multi-layer perceptron (MLP). Under this design, mMTC's true reward-optimal margin under congestion is small by the reward's own calibration, while URLLC's much larger priority weight produces proportionally larger gradients. Because all three slices update the *same* shared parameters, URLLC's abundant, large-magnitude gradients overwrite mMTC's much smaller true signal before it registers. This is a standard shared-parameter/class-imbalance failure mode. We replace the single shared head with *one small MLP per slice*, selected by the context one-hot where the head is applied (Fig. 1(a)). Each slice's decision boundary now has its own parameters, so one slice's gradients can no longer directly overwrite another's.

Algorithm 1 states the training loop, in which the encoder is updated from every request in a sampled minibatch regardless of gNB or slice (centralised training), while action selection at rollout time consults only the acting agent's own embedding row (decentralised execution). Every pending request in a step is its own independent temporal-difference sample, never summed or averaged across a step before the loss is computed, matching our existing single-agent DQN's own granularity.

4.2 Federated Training with Differential Privacy

Fig. 1(b) shows the federated variant, in which each gNB becomes an FL client with a private copy of the same network, training only on its own requests. Transitions are partitioned by owning agent, so no client sees another client's raw admission data, while its GAT encoder still consumes the full joint node-feature snapshot every arm observes. Every R local steps, the server averages all clients' online weights (FedAvg [11]) and broadcasts the result back. We also implement FedProx [10] (a proximal term pulling local weights toward the last broadcast global model), motivated by an implicit, seeded per-gNB load multiplier this project's own cluster-scaling investigation (Section 5.4) later revealed as genuine, uncontrolled client heterogeneity. Swept at $\mu \in \{0.01, 0.1, 1.0\}$ under both a homogeneous-load control and this heterogeneity, FedProx earns no measurable dividend at any tested strength, traced to per-round client drift being negligible relative to the loss scale.

Differential privacy is applied as a standard DP-SGD [1] step, in which every client's gradient is clipped to a fixed norm and, when enabled, perturbed with Gaussian noise scaled by a noise multiplier σ, before that client's own optimiser step. We report σ as the primary privacy parameter, alongside a formal (ε, δ) bound computed via zero-concentrated differential privacy (zCDP) [4] composition of the Gaussian mechanism ($\rho = k/(2\sigma^2)$ over k per-client releases, $\varepsilon(\delta) = \rho + 2\sqrt{\rho \ln(1/\delta)}$). This is a conservative, non-subsampled bound, reported alongside σ rather than in place of it. Algorithm 2 states one federated round precisely. Clipping is applied identically whether or not noise is added, so a σ=0 run and a σ>0 run differ only in whether noise is injected. This is the isolation the privacy-cost comparison (Section 5.2) depends on.

Algorithm 1 GAT-CTDE: centralised training, decentralised execution

Require: shared GAT encoder f_θ, per-slice Q-heads $\{Q_{\phi_k}\}_{k=1}^{K}$, target networks $f_{\theta^-}, \{Q_{\phi_k^-}\}$, replay buffer $\mathcal{D}$, fully-connected adjacency A, discount γ, target-sync period C episodes

for episode $= 1, 2, \ldots$ **do**
  reset environment; observe joint node features $X_1 \in \mathbb{R}^{N \times d}$
  for step $t = 1, \ldots, T$ **do**
    $E_t \leftarrow f_\theta(X_t, A)$ {one embedding row per gNB; centralised: sees every node}
    for each pending request r at gNB i, slice k **do**
      $a_r \leftarrow \arg\max_a Q_{\phi_k}(E_t[i])[a]$ w.p. $1-\epsilon$, else uniform random {decentralised: own row $E_t[i]$ only}
    end for
    $X_{t+1}, \rho_t, \text{done} \leftarrow \text{ENV.STEP}(\{a_r\})$
    store $\big(X_t, X_{t+1}, \{(i, k, a_r)\}_r, \rho_t, \text{done}\big)$ in $\mathcal{D}$
    if $|\mathcal{D}| \geq$ warmup size **then**
      sample minibatch $B \subset \mathcal{D}$
      $E_B \leftarrow f_\theta(X_B, A)$; $E'_B \leftarrow f_{\theta^-}(X'_B, A)$ {target net, no grad}
      for each request $(i, k, a) \in B$ **do**
        $q \leftarrow Q_{\phi_k}(E_B[i])[a]$
        $y \leftarrow \rho + \gamma \cdot (1-\text{done}) \cdot \max_{a'} Q_{\phi_k^-}(E'_B[i])[a']$
      end for
      $\mathcal{L} \leftarrow$ mean Huber loss over every request's (q, y) pair {per-request, never summed per step}
      update θ, $\{\phi_k\}$ by $\nabla\mathcal{L}$; clip grad-norm to 1.0
    end if
  end for
  if episode mod $C = 0$ **then**
    $\theta^- \leftarrow \theta$; $\phi_k^- \leftarrow \phi_k \ \forall k$
  end if
end for

4.3 Experimental Setup

Both the centralised and federated campaigns use 300 training and 50 evaluation episodes per seed. The centralised campaign uses a fixed 30-seed list (seeds 900–929) with three arms: GAT-CTDE, an independent-DQN ablation (one per-gNB DQN instance, no parameter sharing, no topology information) that isolates the GAT/centralised- training contribution, and a single-agent DQN baseline run against the flattened joint state. All three arms use matched hyperparameters (discount factor $\gamma = 0.95$, per-episode epsilon decay). The federated campaign uses the first 10 of these 30 seeds and a five-level noise-multiplier sweep $\sigma \in \{0, 0.5, 1, 2, 4\}$, so its σ=0 level is directly comparable to the centralised campaign's same 10 seeds. All confidence intervals (CIs) are 95% bootstrap percentile intervals (10,000 resamples). Paired comparisons use the Wilcoxon signed-rank test.

Algorithm 2 Federated GAT-CTDE round with DP-SGD

Require: N clients, each with local network $(\theta_i, \{\phi_{i,k}\})$; global snapshot $\bar{\theta}$; local steps per round R; clip norm C_{dp}; noise multiplier σ; FedProx weight μ ($= 0$ unless noted)

for round $= 1, 2, \ldots$ **do**
  for local step $= 1, \ldots, R$, each client i independently **do**
    sample minibatch B_i of i's own requests from i's local buffer
    $E \leftarrow f_{\theta_i}(X_{B_i}, A)$ {full joint state; client's OWN local weights only}
    $\mathcal{L}_i \leftarrow$ per-request Huber TD loss (Alg. 1, lines 11–13) $+ \frac{\mu}{2}\|\theta_i - \bar{\theta}\|^2$ {FedProx term}
    $g_i \leftarrow \nabla_{\theta_i, \phi_i} \mathcal{L}_i$
    if $\sigma > 0$ **then**
      $g_i \leftarrow \text{clip}(g_i, C_{\text{dp}})$; $g_i \leftarrow g_i + \mathcal{N}(0, (\sigma C_{\text{dp}})^2 I)$ {DP-SGD [1]}
    else
      $g_i \leftarrow \text{clip}(g_i, C_{\text{dp}})$ {same clip norm, no noise, isolating the privacy cost}
    end if
    $(\theta_i, \phi_i) \leftarrow (\theta_i, \phi_i) - \eta\, g_i$
  end for
  $\bar{\theta}, \bar{\phi} \leftarrow \frac{1}{N} \sum_{i=1}^{N} (\theta_i, \phi_i)$ {FedAvg [11], equal client weights}
  for each client $i = 1, \ldots, N$ **do**
    $(\theta_i, \phi_i) \leftarrow (\bar{\theta}, \bar{\phi})$ {broadcast}
  end for
end for

4.4 Evaluation Safeguards

This paper's own evaluation pipeline hid four distinct failures a standard pipeline would have missed. Table 1 summarises each against the safeguard that caught it: the training collapse diagnosed in Section 4.1 above, two more diagnosed here, and the fourth in the live deployment (Section 5.5). Two correctness-aware metrics, computed from data every run already logs, let us tell a genuine training failure apart from a legitimate one: mean reward per step, the actual training objective, and block precision, the fraction of blocked requests correctly targeting the slice the reward calibration makes reject-optimal (Section 4.4 states why these are necessary in full).

A separate, unrelated append-mode logging bug had silently stacked three retraining runs' evaluation episodes together in every one of the 30 seeds' logs, hiding a three-way average behind what looked like a single run's result. Once caught, the logs were cleared and evaluation re-run once against the already-trained checkpoints, with no retraining needed. All numbers here reflect that corrected data.

The fixes described in Section 4.1 make the architecture more correct while making `sla_compliance_all_slices` score *worse* (Section 5.1), since blocking mMTC under congestion, the reward-optimal action, never recovers mMTC's own SLA margin in this stress regime. We therefore report two further metrics, both from data every arm already logs: **mean reward per step**, the actual training objective, and **block precision**, the fraction of blocked requests targeting mMTC specifically, the only

Table 1 Four evaluation-pipeline failures this paper's safeguards caught, none of which a standard pipeline would have surfaced.

Failure	Missed by	Caught by	Outcome
Training collapse (§4.1)	Compliance blind to no-op	Correctness-aware pair	0/30→21/30 differentiate
Eval-log contamination	Append-mode log stacking	Episode-index audit	Logs cleared, no retrain
Reproducibility gap (§5.3)	Single-sample overclaim	Repro-vs-replication	Churn claim retracted
Scheduler-floor bug (§5.5)	Can't see cross-layer floor	Direct state instrumentation	>100k readings clear

slice the reward calibration ever makes reject-optimal (undefined for seeds that never block anything). Any evaluation where correct behaviour cannot recover an already-stressed slice's compliance should report a decision-level correctness metric alongside an outcome-level compliance one, the general lesson this collapse makes concrete.

This disagreement is not unique to the one collapse diagnosed in Section 4.1. Re-checking every paired comparison already reported in this paper's centralised (Section 5.1), federated (Section 5.2), disruption (Section 5.3), and cluster-scaling (Section 5.4) campaigns against both metrics together, the correctness-aware pair and `sla_compliance_all_slices` disagree on direction or significance in 44 of 49 comparisons (89.8%) on the seed ranges already committed at the time, independently confirmed at 21/24 (87.5%) on a disjoint supplementary seed range drawn specifically to check whether the first count was itself a one-off. `sla_compliance_all_slices` is not an occasionally-misleading proxy for this problem. On this evidence it is close to uninformative about decision quality on its own, and every number in this paper that matters is reported through the correctness-aware pair, not the compliance metric, for exactly this reason.

The second safeguard this paper applies throughout is a reproduction-versus-replication discipline: every headline paired comparison is recomputed on an independent, disjoint seed sample drawn after the first, and both figures are reported together rather than only the first. This is not a formality. Applied to the disruption campaign (Section 5.3), it directly contradicted an initial reading that independent DQN was architecturally immune to agent churn, a claim that looked confirmed on the committed sample alone and was retracted once the independent sample showed a significant, larger-than-coordinated cost in the opposite direction. Section 5.3 reports the full tally of every headline comparison checked this way, and the churn retraction is the one exception among them, not the rule.

5 Results

5.1 Centralised

The $N = 3$ fully-connected topology used throughout this campaign is deliberately the setting where a shared graph-attention encoder has the least topology to exploit, since adjacency across three gNBs is close to trivial at full connectivity. This makes

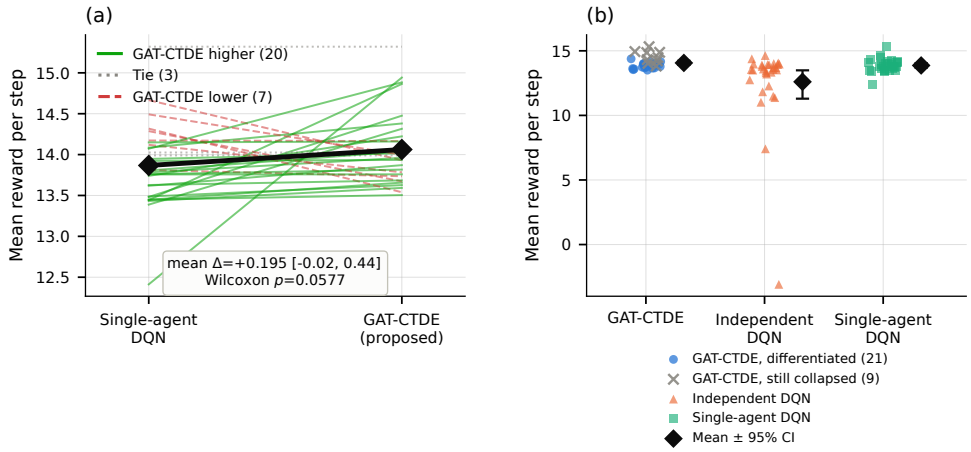


Fig. 2 M2 centralised campaign, per-seed evidence for the correctness-aware metrics (Section 4.4). (a) Paired per-seed comparison against single-agent DQN on mean reward per step. Each line is one seed, coloured by whether GAT-CTDE scored higher (green, 20), tied (grey, 3), or lower (red, 7), the source of this figure's borderline $p = 0.0577$. (b) Full per-seed reward distribution for all three arms. GAT-CTDE is split into seeds that learned genuine differentiated shedding (blue, 21) versus seeds still stuck at the always-accept collapse (grey ×, 9), shown without claiming the split predicts reward, since several still-collapsed seeds score among the arm's highest.

it a clean substrate for the evaluation-integrity findings that follow, uncomplicated by any topology effect. The encoder's coordination advantage is put to a genuine test only once the cluster scales to $N = 19$ under sparser adjacency in Section 5.4.

Table 2 and Fig. 2 report the final, post-fix 30-seed campaign. On `sla_compliance_all_slices`, GAT-CTDE's mean is 0.151, below its own pre-fix number (0.353) since the metric penalises newly-correct mMTC-blocking decisions (Section 4.1), and the paired comparison against single-agent DQN shows no clear direction, at +0.036 (95% CI [−0.050, 0.126], Wilcoxon $p = 0.4549$, 11 wins / 9 ties / 10 losses). On mean reward per step, the metric that reflects the actual objective, GAT-CTDE beats the independent-DQN ablation decisively, by +1.461 (95% CI [0.550, 2.782], $p < 0.0001$, 27/0/3), isolating a real contribution from the shared GAT encoder and centralised training, but its edge over single-agent DQN is small and not statistically decisive at this sample size, at +0.195 (95% CI [−0.017, 0.442], $p = 0.0577$, 20/3/7). Its block precision (0.952) is high regardless, since when GAT-CTDE blocks a request, it is almost always the reward-correct slice, a real, substantial behavioural improvement (Section 4.1) that does not, at this sample size, translate into a decisive reward edge over the single-agent baseline specifically.

5.2 Federated Training and DP

Comparing the centralised campaign against the federated variant with no DP noise, on the same 10 seeds, mean reward per step moves from 14.073 to 13.940, a change of +0.133 (95% CI [−0.138, 0.470], Wilcoxon $p = 0.8125$). This is no measurable cost

Table 2 M2 campaign results, 30 seeds/arm, 95% bootstrap CIs.

Arm	Mean reward/step	Block precision	Compliance*
GAT-CTDE	**14.062** [13.91, 14.23]	0.952	0.151 [0.083, 0.225]
Indep. DQN	12.601 [11.30, 13.49]	0.901	0.049 [0.013, 0.093]
Single-agent	13.867 [13.69, 14.05]	0.987	0.115 [0.061, 0.176]

* Low compliance is the metric artifact Section 4.4 explains, not a performance failure.

Table 3 Training hyperparameters, all fixed across seeds.

Hyperparameter	Value
Learning rate (GAT-CTDE)	1×10^{-4}
Learning rate (DQN baselines)	1×10^{-3}
Discount γ	0.95
Epsilon: start $\rightarrow$ end (decay/episode)	$1.0 \rightarrow 0.05$ (0.985)
Target-network sync	every 10 episodes
Optimiser	Adam
Training / evaluation episodes	300 / 50
GAT encoder (GAT-CTDE only)	
Hidden / output dim	16 / 16
Attention heads	4
Encoder layers	2
Q-head hidden dim	32
Federated + DP (federated GAT-CTDE only)	
Local steps per round	50
Aggregator	FedAvg (μ=0)
DP gradient clip norm	1.0
DP noise multiplier sweep σ	$\{0, 0.5, 1, 2, 4\}$

to federating at this sample size, and both arms' block precision is perfect (1.000) on seeds that block anything at all. Sweeping the DP noise multiplier (Fig. 3(b)), block precision is perfect through $\sigma = 1.0$ then drops sharply, to 0.834 at $\sigma = 2.0$ and 0.584 at $\sigma = 4.0$. This is a real threshold, not a smooth degradation, while mean reward per step stays in a comparatively narrow band across the whole sweep (12.45–14.19), since DP noise erodes *which* slice gets blocked, not whether the policy tries to differentiate at all. The compliance-based version of this same sweep (not shown) is non-monotonic with sign flips between adjacent σ levels. This is a second instance, independent of Section 4.1's architectural finding, of `sla_compliance_all_slices` failing to track the quantity we actually care about. Read together, federation and the FedProx null result (Section 4) answer a deployability question directly: periodic, infrequent synchronisation is sufficient for this architecture, so a disaggregated RAN Intelligent Controller does not need per-step coordination across gNBs to match centralised training's own reward, and differential privacy can be added at a bounded, known precision cost rather than an unpredictable one.

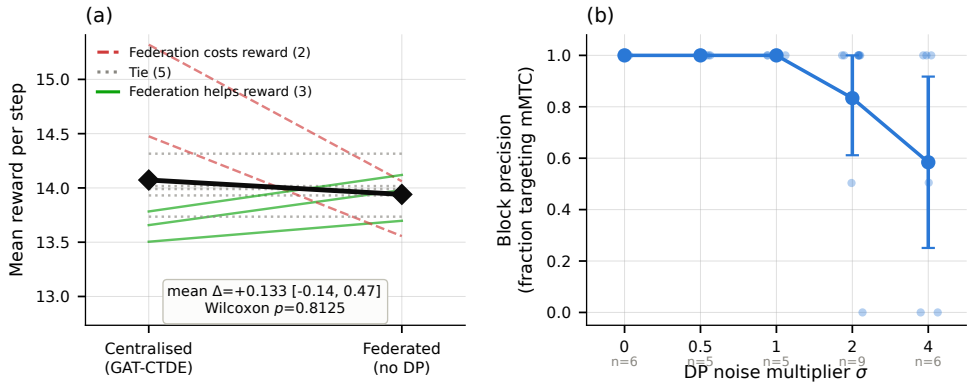


Fig. 3 M3 federated and differential-privacy sweep, per-seed evidence. (a) Paired per-seed federation cost on mean reward per step, same 10 seeds. Centralised scores higher on 2 seeds, federated on 3, 5 ties, no measurable difference (Wilcoxon $p = 0.8125$). (b) Privacy-utility curve using block precision against the DP noise multiplier σ, individual seed values (light dots) behind the mean, with the seed count that ever blocked anything at each σ printed below (precision is undefined otherwise). Precision is perfect through $\sigma = 1.0$, dropping sharply by $\sigma = 4.0$.

5.3 Disruption

We evaluate every arm's already-trained, frozen checkpoint (no retraining) against three mid-episode disruptions injected at evaluation time. *gNB dropout* forces that gNB's requests to reject and zeroes its row in the joint feature matrix for a window, so a policy cannot simply learn around the corrupted feature. *Demand spike* is a transient arrival-rate multiplier for a fixed window, entirely external to the policy, acting on the arrival process the environment synthesises for the following step. *Agent churn* substitutes that gNB's decisions with a fresh, never-trained copy of its own policy class for a window, leaving every other agent's decisions untouched, a different, well-motivated failure mode from dropout (active-but-uncoordinated vs. absent) that applies to every genuinely multi-agent arm. Single-agent DQN is excluded from it, and all three disruption kinds are evaluation-only perturbations that never update the policy itself. Severity is the window's duration (10%/30%/60% of the episode) for dropout and churn, and the arrival-rate multiplier (2×/4×/8×) for spike. We normalise reward by each step's own pending-request count before averaging, since raw reward mechanically inflates with request volume regardless of decision quality.

Table 4 reports each arm's own per-severity dropout and churn cost where an individual triple was computed. Every cost accelerates with severity rather than growing proportionally (Wilcoxon $p \leq 0.03$, 9 or 10 of 10 seeds hurt in every case). Demand spike does not corrupt decision quality on the volume-invariant block-precision metric (Fig. 4(c)); its apparent per-step reward gain is a denominator artifact (diluted by request volume), not better decision-making.

The federated arm's own churn cost above was regenerated post-hoc from the still-intact frozen checkpoint after the original per-seed logs were found missing during this paper's own reproducibility audit, with environment drift over the intervening months

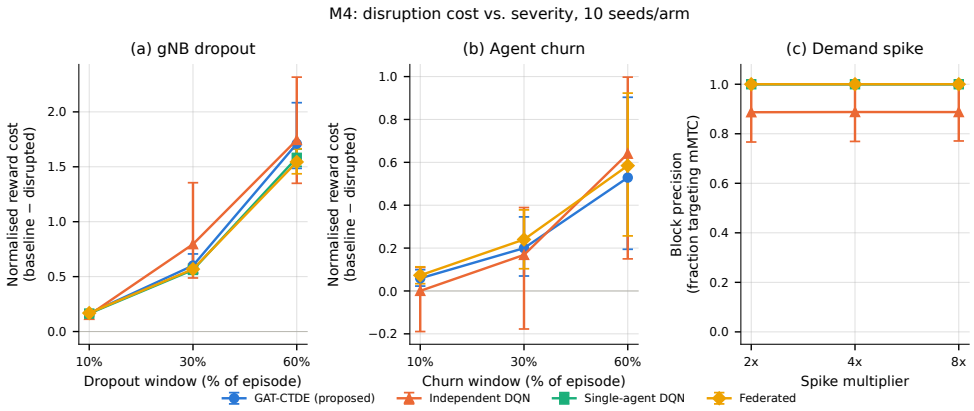


Fig. 4 M4 disruption campaign, 10 seeds/arm. (a)-(b) Normalised reward cost (baseline − disrupted) vs. severity for dropout and agent churn. Every arm, every severity, shows a real cost that accelerates with severity rather than growing linearly. (c) Block precision vs. demand-spike multiplier: flat and near-ceiling for every arm, unlike (a)-(b), since spike does not threshold on the metric that matters.

Table 4 Disruption cost by severity, committed sample (900–909), normalised reward. †: not individually computed beyond the qualitative accelerating pattern common to every arm (see text). ‡: agent churn does not apply to a single-agent architecture.

Arm	Dropout (10/30/60%)	Churn (10/30/60%)
GAT-CTDE	+0.165/+0.601/+1.710	+0.059/+0.200/+0.529
Federated (σ=0)	accelerates†	+0.078/+0.246/+0.585
Independent DQN	accelerates†	+0.000/+0.168/+0.640
Single-agent DQN	accelerates†	‡

identified as the cause by systematic elimination of every code- and data-level alternative. Independent DQN's own churn cost needs both samples reported together to state honestly: the committed value above is small and not conventionally significant ($p = 0.084/0.106/0.065$), which an earlier pass read as independent DQN being architecturally immune to churn, since it never attends to another agent's features. An independent-seed replication (Table 5) contradicts that reading directly, significant at every severity and *larger* than GAT-CTDE's or the federated arm's at matched severities, the opposite of immune. We do not conclude independent DQN is immune to churn, since the direction was consistent across both samples (churn always hurts, never helps); what was wrong was treating a borderline, underpowered $n = 10$ result as a confirmed architectural property, and this section reports the corrected reading, not the retracted one.

Table 5 reports this paper's four cross-section headline comparisons against their own independent-seed replication; GAT-CTDE's own dropout costs also replicate closely (+0.170/+0.552/+1.449, against the committed sample's

Table 5 Independent-seed replication (1000–1009) against the committed sample (900–909) for this paper's four cross-section headline comparisons. †: the one comparison that does not replicate.

Comparison	Committed (900–909)	Replication (1000–1009)
GAT-CTDE − indep. DQN (reward)	+1.461 [0.550, 2.782], $p < 0.0001$	+0.767 [0.391, 1.233], $p < 0.0001$
GAT-CTDE − single-agent (reward)	+0.195 [−0.017, 0.442], $p = 0.058$	+0.445 [−0.049, 1.280], $p = 0.111$
Federation cost (reward)	+0.133 [−0.138, 0.470], n.s.	−0.080 [−0.241, 0.001], n.s.
Independent-DQN churn cost†	+0.000/+0.168/+0.640, n.s.	+0.097/+0.657/+1.223, $p = 0.0020$

+0.165/+0.601/+1.710). Three of the four hold up with overlapping confidence intervals across samples; the fourth, independent DQN's churn cost (flagged †), does not, and is the retraction above.

5.4 Cluster-Size Scaling

We extend the cluster from $N = 3$ to $N \in \{7, 19\}$ under three adjacency structures (fully-connected, ring, hex-grid), scaling synthetic arrivals with N to hold offered load per gNB roughly constant. GAT-CTDE is the only arm whose Q-head consumes this adjacency. Independent DQN and single-agent DQN never read a graph structure and serve as a topology-invariance check. We normalise reward per gNB before averaging (the same fix Section 5.3 applies on the severity axis) and report block precision alongside, since `sla_compliance_all_slices` is OR-aggregated across gNBs and trends toward 0 as N grows regardless of policy quality. This is a third instance of this paper's central evaluation-integrity finding, now on the cluster-size axis. The primary sample is 12 seeds (900–911) at $N = 19$, all three topologies, all three arms. An independent-seed replication (1000–1002, disjoint) checks generalisation.

At the full 12-seed sample, the paired per-gNB reward-margin comparison at $N = 19$ is not significant for any topology (fully-connected −0.175 [95% CI −0.528, +0.076], $p = 0.7422$; ring −0.704 [−1.879, +0.042], $p = 0.7422$; hex −0.695 [−1.873, +0.053], $p = 0.9102$). This is mechanically explicable, not a red flag, since single-agent DQN's always-accept collapse (below) collects the reward's service term on every request with no rejection cost, while GAT-CTDE pays a real cost every time it correctly rejects mMTC. This paper's evidence for GAT-CTDE's advantage at $N = 19$ lives in collapse resistance and block precision, not the reward-margin metric.

Single-agent DQN collapses totally at $N = 19$: 12/12 primary-sample cells and 3/3 replication cells, every topology, zero exceptions (15/15 combined). This is confirmed at the individual-decision level by a direct Q-value probe identical in method to Section 4.1's original diagnosis. GAT-CTDE collapses only partially, and independent DQN never fully collapses (0/36) but never cleanly differentiates either, sitting in a narrow, mediocre 0.29–0.49 block-precision band identical across all three topologies, as expected for an arm that never consumes adjacency (the N=7 extension below tests this same claim again under an independent replication sample and finds a small,

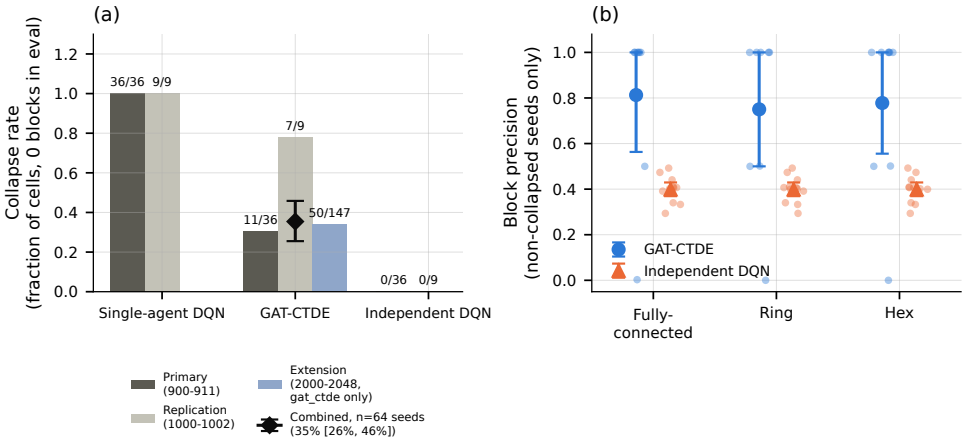


Fig. 5 $N = 19$ cluster, per-seed evidence across independent seed samples. (a) Collapse rate (fraction of eval runs with zero blocks), pooled across all three topologies. Single-agent DQN collapses totally in every sample. GAT-CTDE's rate varies sample-to-sample (per-sample breakdown in the body text), converging to a combined, seed-level-bootstrapped estimate of 35.4% [25.5%, 45.8%] across all 64 independent seeds (diamond marker, error bar). (b) Block precision by topology, primary sample, non-collapsed seeds only (single-agent DQN has none to show). GAT-CTDE's non-collapsed seeds are mostly near-ceiling but with a low-precision seed at every topology, while independent DQN sits in a narrow, mediocre band, identical across topologies as expected for an arm that never consumes adjacency.

rare exception to "never"). GAT-CTDE's own non-collapsed seeds show a topology-dependent secondary failure mode, since most hold near-perfect precision, but at least one seed per topology drops to 0.0–0.5, including one seed showing perfect precision (1.000) at fully-connected and near-zero (0.000) at ring/hex on the identical seed, environment stream, and initial weights.

GAT-CTDE's own collapse rate initially disagreed sharply between the primary sample (31%) and the replication sample (78%). A third, independent sample (49 seeds, disjoint from both) resolved rather than left the range open, pooling to 68/192 cells collapsed, 35.4% (seed-level bootstrap 95% CI [25.5%, 45.8%] across the 64 independent seeds, since collapse status correlates within a seed across its three topologies). None of the three samples was wrong: each was an honest reading of a genuinely wide distribution, and no single small sample should be trusted alone.

At N=7, the original pilot was a 3-seed sample, never resampled at scale because N=19's collapse question was judged the higher-value target: GAT-CTDE and single-agent DQN both held 1.000 precision in that pilot (0/3 collapsed each). We closed this gap with a properly powered resample under the identical, unmodified stress environment (`ClosedLoopKpmSource`, no recalibration of any kind), 12 seeds primary (900–911, the same primary identity as above) plus an independent 12-seed replication (1000–1011, disjoint), three topologies, three arms, the same 100-train/20-eval budget the original pilot used.

Table 6 N=7 collapse rate, clean stress environment (`ClosedLoopKpmSource`, unmodified), 12 seeds/sample, 3 topologies × 3 arms, 95% bootstrap CI.

Arm	Primary (900–911)	Replication (1000–1011)	Original 3-seed pilot
Single-agent DQN	91.7% [75.0, 100]	75.0% [50.0, 100]	0% (0/3)
GAT-CTDE	41.7% [16.7, 66.7]	58.3% [33.3, 83.3]	0% (0/3)
Independent DQN	0.0% [0.0, 0.0]	16.7% [0.0, 41.7]	0% (0/3)

Table 6 reports both samples directly, and the pilot's finding does not survive: single-agent DQN's clean N=7 collapse rate is far above the pilot's 0%, the single clearest result in this scaling extension, not merely a wider interval but the opposite conclusion from a properly powered sample. GAT-CTDE's N=7 rate is statistically indistinguishable from its own N=19 pooled estimate (35.4% [25.5%, 45.8%] above), meaning its collapse-proneness is present from N=7 onward rather than an effect that only emerges at large cluster size. Independent DQN's primary sample matches the N=19 pooled reading exactly (never collapses), but its replication sample does not, the same small, real, non-zero exception the N=19 result above already hints at, now observed independently at a second scale. Its own "never fully collapses" claim is best read as "rarely, not never" at both scales, not retracted at either. This closes the one gap the N=19 pilot comparison flagged (an under-powered pilot, not a properly-sampled result), using the same environment throughout, with no recalibration of any kind.

5.5 Live Deployment

Every multi-gNB, federated, and disruption result above is offline (Section 3). This section reports what a real OAI testbed can and cannot establish about the offline-trained policy, using a single-agent-DQN checkpoint (seed 900, dimension-matched to the live evaluation path) against a real gNB, real UEs, and real per-slice User Datagram Protocol (UDP) traffic, at native load (1 UE/slice) and double native load (2 UEs/slice).

An early live campaign on this rig read findings from data collected while a scheduler-configuration bug was active: `saclb_live.yaml`'s per-slice PRB-ratio ranges (commits 82b20f6 to 40ae2f3) converted to raw PRB grants below the gNB scheduler's hard 5-PRB minimum, a cross-layer interaction a config-level review alone could not have caught, producing catastrophic Radio Link Control (RLC) retransmission failures indistinguishable from genuine per-slice capacity scarcity. Fixed at its root (commit cda9d65, rescaling the config's ratio ranges to clear the floor), every result below was collected after the fix, with the scheduler's own state directly instrumented and confirmed clear of the floor throughout (zero below-floor grants across more than 100,000 combined readings).

The re-collected picture is a genuine deployability result, not an admission-control result. A live capacity probe at a deliberately wide-open ceiling found all three slices serve essentially 100% of offered demand with 0.000% loss, at both 1 and 2 UEs per slice (URLLC 100%, eMBB ≈99.9–100%, mMTC ≈105–107% of its

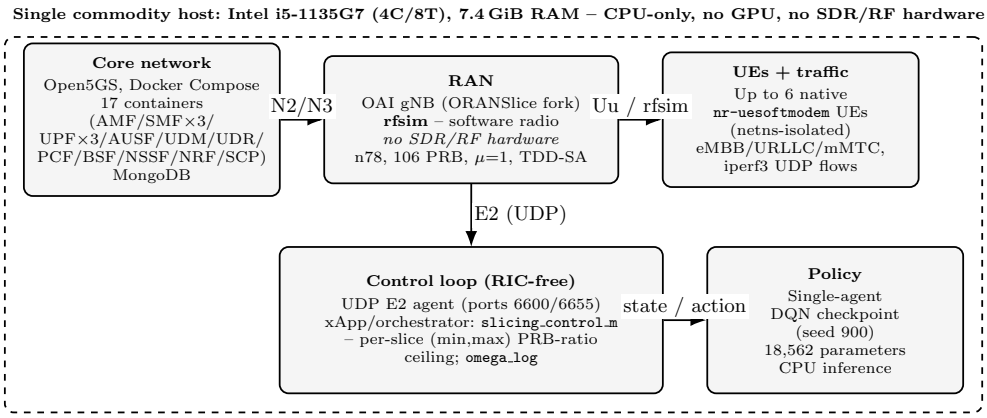


Fig. 6 This rig's real testbed stack: a single-host, software-radio (`rfsim`), RIC-free O-RAN deployment, distinguished here from SDR/hardware-in-the-loop O-RAN testbeds. Every block and label is a directly measured or configured value from this rig (no component is borrowed from any reference testbed figure). All five blocks run as processes/containers on the one physical host outlined by the dashed boundary; no element of this stack is external hardware.

own bursty-average demand). This rig has never been shown to reach a load where any slice's admission decision carries a genuine capacity consequence. The frozen checkpoint's decision logic executes correctly end to end at native load: real traffic through the real E2 loop, real accept/reject decisions, its characteristic low-priority-only blocking (mMTC targeted, URLLC/eMBB untouched) reproducing at both loads with no catastrophic failure. But since the environment never reaches scarcity, this pattern is at least as consistent with the policy's own trained priority bias (mMTC's lowest `priority_weight`, Section 5.1) as with contention management. **The honest claim this rig supports is deployability, not admission-control-under-contention**, which is instead tested in the disclosed offline stress environment (Sections 5.1–5.3, 5.4), deliberately calibrated harder than this rig has ever been measured to reach.

A separate, standing finding is unaffected by this re-framing: a unit-mismatch defect in the live SLA-margin diagnostic (a per-slice backlog reading divided by a calibration constant tuned for the offline environment's synthetic proxy, rather than the real Medium Access Control (MAC) scheduler's byte-count scale) is real regardless of load, and is fixed by converting backlog to a queueing-delay estimate via Little's Law against each slice's own latency budget. An earlier recalibration was retired once the floor fix removed the collapse it had targeted.

6 Conclusion

We set out to test whether a standard evaluation metric, and a live testbed's own evidentiary reach, can be trusted in multi-agent, privacy-constrained, disruption-prone slice admission control, and found neither can go unchecked. A standard SLA-compliance metric, an unguarded append-only log, an unseeded weight-initialisation path, and a cross-layer scheduler-configuration bug together hid four distinct failures:

a training collapse the compliance metric scored no worse than correct behaviour, a hidden eval-data contamination bug, a same-seed-only reproducibility gap, and a live scheduler-floor misconfiguration that had silently manufactured the appearance of genuine contention in earlier live data. None of the four would have been caught by a standard pipeline without this paper's correctness-aware metrics, reproduction-replication discipline, and direct live instrumentation of the scheduler's own state.

GAT-CTDE, the multi-agent architecture that served as this evaluation's substrate, delivers a decisive, statistically significant improvement over an independent-learner ablation on the correctness-aware reward metric, though its edge over our existing single-agent baseline specifically is small and not decisive at this sample size. This is an honest, non-inflated architectural finding, not the headline result this paper argues evaluation practice should not be built around alone.

Extending to federated training with differential privacy, federation adds no measurable reward cost, privacy is costly only past a real precision threshold, and FedProx earns no measurable dividend under genuine per-gNB load heterogeneity at any tested strength. Under mid-episode gNB dropout, demand spikes, and agent churn, dropout and churn produce a real, accelerating cost with severity while demand spikes do not threshold at all on the metric that matters. Cross-checking every result against an independent seed sample, rather than the same seeds reproduced again, caught and retracted one such claim (that independent DQN is immune to agent churn) that a single sample had made look confirmed but was not.

Scaling the cluster to 19 gNBs, single-agent DQN's collapse becomes total and GAT-CTDE's remains only partial. GAT-CTDE's own collapse rate, read at two disagreeing values across two small samples, was resolved with a third, independently-drawn sample, converging on 35.4% [25.5%, 45.8%] across 64 seeds, a rate already present at N=7 rather than an N=19-specific phenomenon. Finally, a live single-gNB deployment establishes what this rig can and cannot show: the offline-trained decision logic executes correctly end to end, a genuine deployability result, but a cross-layer scheduler-configuration bug had earlier manufactured the appearance of live contention this rig cannot actually produce. Fixed at its root, this rig has never been shown to reach a load at which any slice's admission decision carries a real capacity consequence, so admission-control-under-contention remains a claim this paper makes only in the disclosed offline stress regime, not live.

Four limits bound every claim in this paper. First, every multi-gNB, federated, and disruption result is offline-only, since this rig cannot yet sustain a live multi-gNB test. Second, the offline stress environment is deliberately calibrated harder than any congestion level this rig has been measured to reach, a disclosed design choice rather than a validated one. Third, the graph representation models only the gNB cluster, with no multi-access edge computing nodes or compute-placement dimension. Fourth, the federated scenario is a synthetic, single-operator privacy setting, motivating periodic synchronisation operationally rather than on a privacy case this scale cannot substantiate. Together, these results argue that admission-control evaluation in multi-agent, privacy-constrained, disruption-prone settings needs metrics tied to the actual training objective, that architectural claims need more than one seed sample before they

are trusted, and that a live testbed's own evidentiary reach needs to be checked, not assumed, before its silence about contention is read as safety.

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Declarations

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- **Competing interests:** The authors declare no competing interests.
- **Ethics approval and consent to participate:** Not applicable, since this study involves no human participants, animal subjects, or identifiable patient data. All reported results are either offline simulation campaigns or measurements from a laboratory radio testbed.
- **Consent for publication:** Not applicable, since this manuscript contains no personal data or identifying information about any individual.
- **Data availability:** The experiment logs, checkpoints, and campaign result files underlying this paper's findings are available from the corresponding author on reasonable request.
- **Materials availability:** Not applicable, since this study involves no physical or biological materials.
- **Code availability:** The framework and experiment scripts used to produce this paper's results, including the reproducibility manifest and analysis scripts used to independently re-derive every table and figure in this paper, are available from the corresponding author on reasonable request.
- **Author contribution:** M. P. Kunasegran: conceptualisation, methodology, software, investigation, writing (original draft). W. L. Pang: supervision, writing (review and editing). S. K. Phang: supervision, writing (review and editing).

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